

Pebblebrook 90-Day Forecast

Two hotels with opposite problems — and \$451K–\$528K sitting on the table.

The Short Version

Boston is nearly sold out for fall and charging the same rate on a Hans Zimmer sellout as on a slow Tuesday.

Santa Monica has 40% of rooms sitting empty for six straight weeks with no plan to fill them.

The forecast found both. The recommendations fix both.

Hotel	Problem	Action	Revenue
Boston	Underselling 18 high-demand nights	Tiered rate increase by occupancy band	~\$193K
Santa Monica	41-night soft window at 59% occupancy	Targeted promotion — act now, guests decide in April–May	~\$258K–\$335K

How the Forecast Works

For each future night, three questions:

1. **What's already booked?** From OTB data — this is fact, not a prediction.
2. **How many more will book before check-in?** Measured from 26 actual booking snapshots in the data.
3. **Is there an event pulling demand?** From events data, scaled by hotel size and capped so one festival doesn't distort the whole forecast.

Add those three up. That's the occupancy forecast. Multiply by rate to get RevPAR.

Forecast Occupancy	=	Already Booked	+	Expected Pickup	+	Event Lift
Forecast RevPAR	=	Forecast Occupancy	×	Current ADR		

Every number traces to a specific row. Nothing is a black box.

What I Found in the Data (Before Running a Single Forecast)

Three problems surfaced immediately after loading the data.

The data was multiplying itself. Joining the events and booking tables naively produced 28,276 rows from two tables that should have produced 180. Each event row was attaching to every booking row for that date — a 6x inflation. Fixed by pre-aggregating events to one row per date. If this had gone unnoticed, every forecast number downstream would have been wrong.

Boston showed rooms sold beyond capacity. 32 date-rows showed occupancy above 100% — more rooms sold than exist. Either real overbooking or a system error. Capped at 100% for forecasting, flagged to operations. This became Recommendation 3.

Revenue math checked out. $ADR \times \text{rooms sold}$ should equal revenue sold. Largest gap: \$1.36. Rounding, not a problem. Accepted.

The forecast only ran after all three cleared.

What the Numbers Show

Boston — nearly full, leaving money on the table

Metric	Value
Average Occupancy	88.7%
Average RevPAR	\$298.91
Nights fully sold out	26 of 90
Peak night	Sep 17 (Hans Zimmer) — \$432 RevPAR

Boston guests book late. The data shows +56% of final occupancy comes in the last 90 days — a steep, fast curve. By the time a promotional rate would matter, the hotel is already full. The lever here is **rate, not volume**.

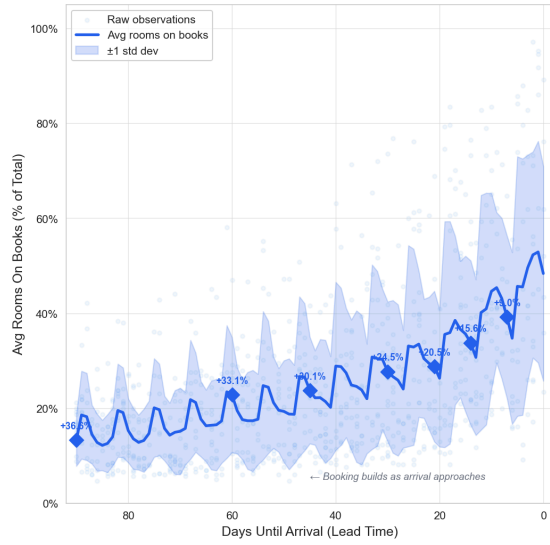
Santa Monica — room to grow, and a deadline to act

Metric	Value
Average Occupancy	62.0%
Average RevPAR	\$157.46
Soft window	Aug 4–Sep 13 (41 nights at 59% avg occupancy)
Nights fully sold out	0 of 90

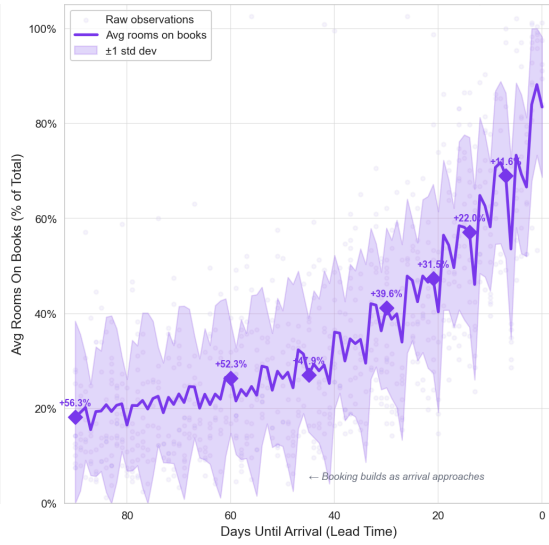
Santa Monica guests book 45–60 days out. That means April and May are the window to move August demand. Wait until July and those travelers have already booked elsewhere.

Historical Pickup Curve — How Bookings Build Toward Arrival
Annotations show avg additional rooms expected to be picked up (as % of total rooms)

HOTEL_A — Santa Monica, CA
Historical Pickup Curve

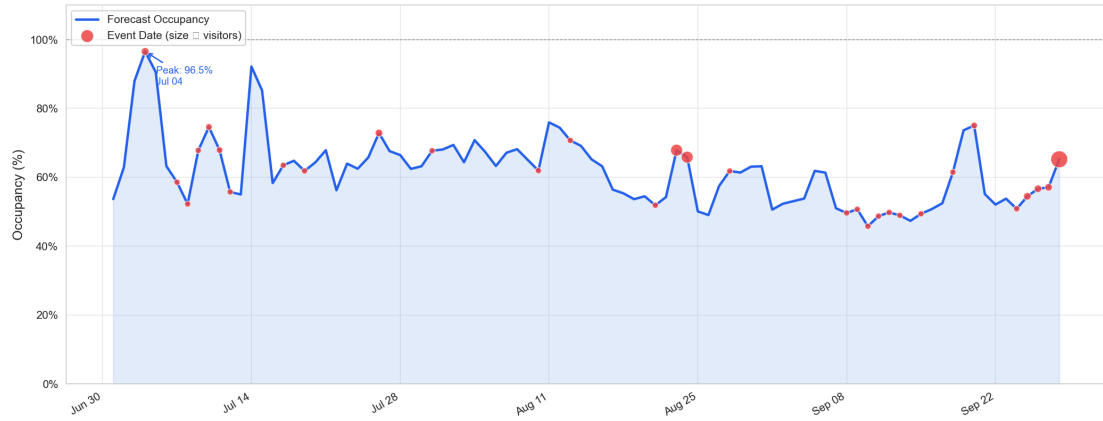


HOTEL_B — Boston, MA
Historical Pickup Curve

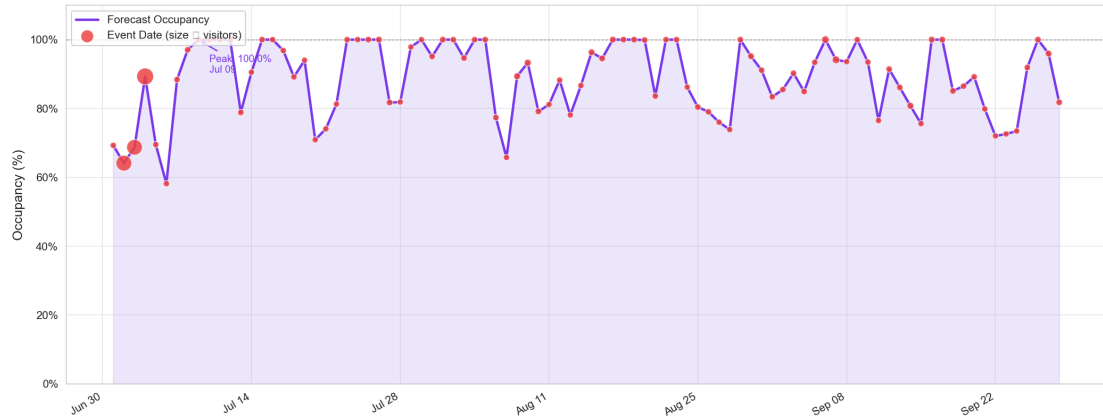


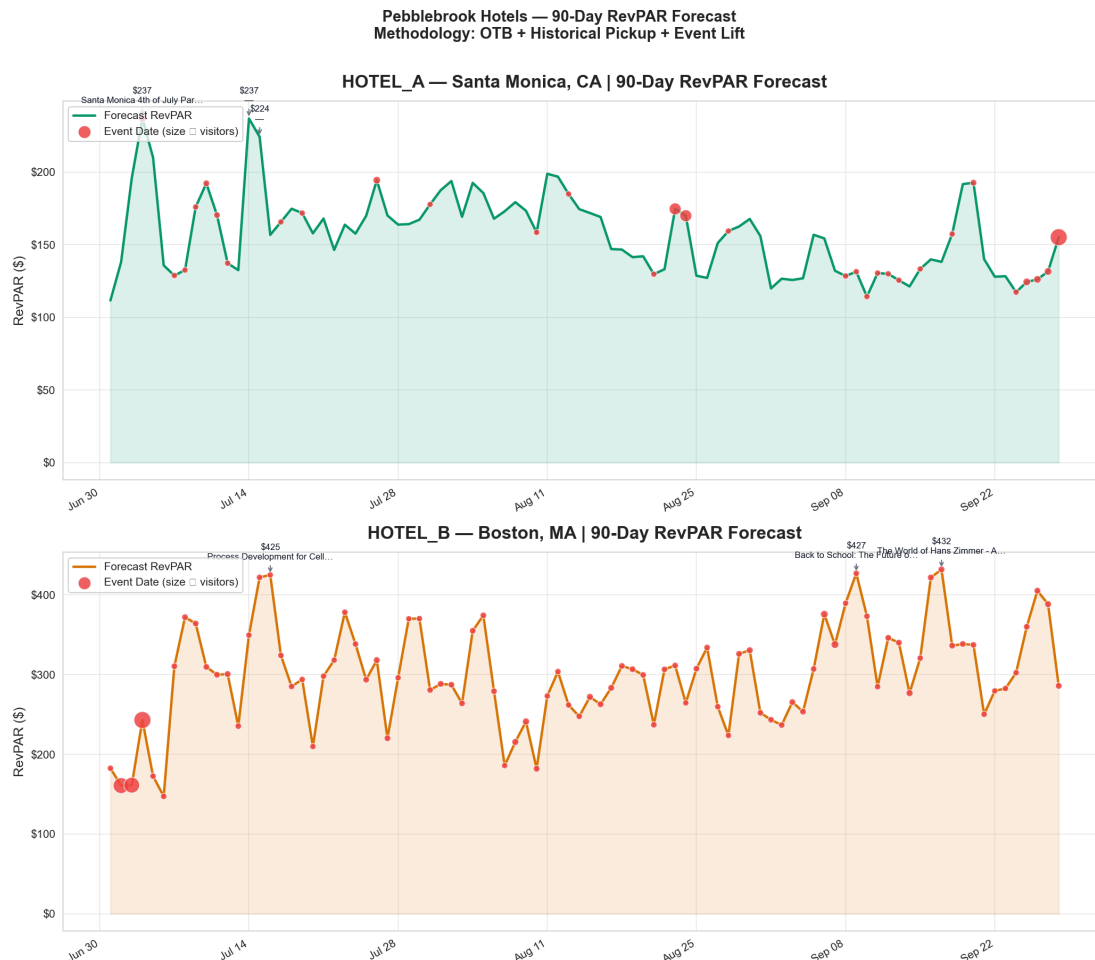
Pebblebrook Hotels — 90-Day Occupancy Forecast
Methodology: OTB + Historical Pickup + Event Lift

HOTEL_A — Santa Monica, CA | 90-Day Occupancy Forecast



HOTEL_B — Boston, MA | 90-Day Occupancy Forecast





Recommendation 1 — Boston Rate Program · ~\$193K

Eighteen nights where the hotel has pricing power it isn't using. Three tiers:

Sellouts — raise rate by \$80 Six nights at 100% occupancy. The room will sell regardless. Rate is the only question.

Date	Current ADR	Driver
Sep 17	\$432	Hans Zimmer
Sep 9	\$427	Translational Digital Pathology Summit
Jul 16	\$425	Process Dev Cell Therapies Summit
Jul 15	\$422	Weird AI @ Boch Center
Sep 16	\$422	LSX World Congress USA
Sep 26	\$405	John Mulaney @ Boch Center

$6 \times 238 \text{ rooms} \times \$80 \approx \$114\text{K}$

Near-sellouts (91–98% occupancy) — raise by \$30 12 nights with strong demand and a few rooms left. A conservative \$30 lift assumes some guests reprice out.

$12 \times 238 \times 0.94 \times \$30 \approx \$80\text{K}$

Watch list (85–89% occupancy) Don't act yet. Check pickup velocity in 7 days. Promote to the tier above if occupancy hits 92%.

Recommendation 2 — Santa Monica Demand Push · ~\$258K–\$335K

Six weeks. No major events. School back in session. Leisure travel drops. The forecast shows 4 in 10 rooms empty every night from Aug 4 to Sep 13.

The window to act is now — Santa Monica guests are making August plans in April and May.

Scenario	Extra Rooms/Night	41 nights × \$258 ADR
Conservative (8 pt occupancy lift)	25 rooms	~\$265K
Optimistic (10 pt lift)	31 rooms	~\$328K

An 8–10 point lift from a targeted soft-period promotion is the low end of what these campaigns typically deliver.

Recommendation 3 — Flag the Boston Overbooking

32 date-rows in April showing occupancy above 100%. The data doesn't tell us whether that's intentional overbooking or a system sync issue — but Operations should know either way.

How I Built This — AI as Co-Pilot, Not Autopilot

I act as the decision-maker; AI agents handle the syntax. Architecture first, code second. Every AI output gets verified against the actual data before anything downstream runs.

Stack (mirrors Python + FastAPI environment):

- **Claude Code (CLI)** — AI co-pilot running in the project workspace
- **@pipeline-architect** — purpose-built agent for data prep, reconciliation, and forecast math
- **@business-strategist** — purpose-built agent for translating model output into revenue actions
- **/reconcile-data** — custom skill that runs as a hard gate between data prep and forecasting
- **CLAUDE.md** — project constraints encoded once, loaded into every agent session: no ML models, modular functions only, every number must be traceable

Encoding constraints upfront means I'm not re-explaining the rules on every prompt. The agent already knows the boundaries before it writes a line.

The loop I ran for every stage:

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SPEC      → Write the business constraint before any code
DECOMPOSE → Break into independent, testable stages
GENERATE  → Delegate boilerplate to the agent (joins, chart scaffolding,
VERIFY    → Check output against the data – not the AI's explanation of
COMMIT    → Only move to the next stage after the check passes

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Stages in practice:

Stage	Agent / Skill	Output	What I verified
 Data Prep & Merge	@pipeline-architect	1_data_prep.py , merged_data.csv (28,276 rows)	Caught fan-out + occupancy outliers + missing event data
 Reconcile / Audit	/reconcile-data skill	1b_data_patch.py , reconciled_data.csv (4,732 rows)	Fan-out resolved, occupancy capped, integrity confirmed
 Forecast	@pipeline-architect	2_forecast_model.py , forecast_90days.csv , 3 charts	Formula audited row-by-row; zero residual
 Translate	@business-strategist	PRESENTATION.md , SUMMARY.md	Every revenue figure cross-checked against CSV

Why verification can't be skipped: The fan-out error — data inflating 6x — wasn't flagged by the AI. It surfaced when I checked row counts manually. The AI wrote the merge code without error; the data problem was invisible to it. Same with the 100%+ occupancy rows. AI is fast and fluent. It doesn't know when it's wrong.

Before any number entered the forecast, it had to clear four checks. Fail any one, the agent rewrites:

- $Occupancy \in [0\%, 100\%]$ on every row
- $rooms_sold + left_to_sell + ooo = total_rooms$ exactly
- $ADR \times rooms_sold \approx revenue_sold$ within \$1.50 tolerance
- Event join must not increase row count

This is the same pattern I built at OMEL AI for enterprise chatbots — intercept output, validate against domain rules, block anything that can't be proven. That layer is what turns "AI-generated" into "production-trustable."

What Would Make This Better

What's missing today	What it would take
Optimal rate recommendation (not just "raise it")	Price elasticity layer + comp-set data
Competitor awareness	OTA comp-rate feed — react when others reprice
Live updates	Replace 26 static snapshots with daily OTB stream
Cancellation response	Already modular — wire to event cancellation feed
Dashboard visibility	Output connects directly to Tableau via FastAPI endpoint